

Math 403 Homework #2, Spring 2025

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(1 point for each of up to 3 collaborators who also list you)

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Due: 11:59pm Saturday 8 February 2025

READING ASSIGNMENTS

/48

7. for Thu. 30 January
 - [Serge Lang, *Linear Algebra*, Chapter XII, §1–§2] separating hyperplanes
 - [Serge Lang, *Linear Algebra*, Chapter XII, §3–§4] support hyperplane, extreme point
8. for Tue. 4 February
 - [Wikipedia, *Grassmannian*] as much as you're willing; thru §3 (“*as a set*”) at least
 - [Wikipedia, *Group action*] get a feel for the concept; no need to overdo it
9. for Thu. 6 February
 - [multivariable calculus, any source]: open & closed sets; differentiability; chain rule $D(f \circ g) = Df \circ Dg$; inverse function theorem: Df invertible $\Rightarrow f$ locally invertible
 - [Wikipedia, *Topological space*] get a feel; through §5.1 (*Metric spaces*), say
 - [Wikipedia, *Differentiable manifold*] read up through §2 (*Definition*)
10. for Tue. 11 February
 - [Lax, 7. Isometry, p.87–89] isometry, orthogonal group
 - [Lax, 7. Complex Euclidean structure, p.95–96] unitary matrices
 - [Treil, §5.6–§5.7] unitary transformations, rigid motions
11. for Thu. 13 February
 - [Treil, §6.1–§6.2] Spectral theorem, normal operators

EXERCISES

1. Find the Jordan form of a 5×5 matrix A whose sole eigenvalue is 3, given that $A - 3I$, $(A - 3I)^2$, $(A - 3I)^3$, and $(A - 3I)^4$ have kernels of dimension 2, 3, 4, and 5, respectively. Do the same thing if the kernel dimensions are 2, 4, 5, and 5, respectively.

Solution

$$\begin{bmatrix} 3 & 0 & 0 & 0 & 0 \\ 0 & 3 & 1 & 0 & 0 \\ 0 & 0 & 3 & 1 & 0 \\ 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 3 \end{bmatrix} \quad (1)$$

Upto permutations of jordan blocks.

$$\begin{bmatrix} 3 & 1 & 0 & 0 & 0 \\ 0 & 3 & 1 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 3 \end{bmatrix} \quad (2)$$

Upto permutations of jordan blocks.

2. Find a Jordan form and a corresponding Jordan basis for the matrix

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$$A = \begin{bmatrix} 7 & 1 & 2 & 2 \\ 1 & 4 & -1 & -1 \\ -2 & 1 & 5 & -1 \\ 1 & 1 & 2 & 8 \end{bmatrix}.$$

Solution

The characteristic polynomial for this matrix is $p(x) = (x - 6)^4$. The geometric multiplicity is 2. Which means that there are 2 jordan blocks. The eigen vectors are.

$$v_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}$$
$$v_2 = \begin{bmatrix} -1 \\ -1 \\ 0 \\ 1 \end{bmatrix}$$

The geometric multiplicities are 2,3 and 4 for $(A - 6I)$, $(A - 6I)^2$, and $(A - 6I)^3$ respectively. So the matrix's jordan form is as below.

$$\begin{bmatrix} 6 & 0 & 0 & 0 \\ 0 & 6 & 1 & 0 \\ 0 & 0 & 6 & 1 \\ 0 & 0 & 0 & 6 \end{bmatrix} \quad (3)$$

3. Let V be a vector space over $\mathbb{R}$ or $\mathbb{C}$ of finite dimension. Let $\varphi : V \hookrightarrow W$ be an injective homomorphism. Show that if μ is a norm on W then $\mu \circ \varphi$ is a norm on V .

Solution

For $\mu \circ \varphi$ to be a norm on V it has to satisfy certain properties of a norm.

1. $\mu \circ \varphi(x) = 0 \Leftrightarrow x = 0$

Since μ is already a norm, $\mu(x) = 0 \Leftrightarrow x = 0$, but φ is injective therefore $\ker(\varphi) = 0$, so $\mu \circ \varphi(x) = 0 \Leftrightarrow x = 0$.

2. $\mu \circ \varphi(\alpha x) = |\alpha| \mu \circ \varphi(x)$ for $\alpha \in$ the field.

Since μ is a norm, then $\mu(\alpha x) = |\alpha| \mu(x)$, and since φ is linear $\varphi(\alpha x) = \alpha \varphi(x)$. $\mu \circ \varphi(\alpha x) = \mu(\alpha \varphi(x)) = |\alpha| \mu \circ \varphi(x)$, satisfying the property.

3. $\mu \circ \varphi(x + y) \leq \mu \circ \varphi(x) + \mu \circ \varphi(y)$ Since μ is a norm $\mu(x + y) \leq \mu(x) + \mu(y)$. So $\mu \circ \varphi(x + y) = \mu(\varphi(x) + \varphi(y)) \leq \mu \circ \varphi(x) + \mu \circ \varphi(y)$.

So $\mu \circ \varphi$ is a norm on V .

4. Let ν be a norm on $V = \mathbb{R}^n$ or $\mathbb{C}^n$ with unit ν -ball $B_\nu = \{\mathbf{x} \in V \mid \nu(\mathbf{x}) \leq 1\}$. Prove [/3](#) that B_ν is closed, bounded, convex, and *equilibrated*: $\mathbf{x} \in B_\nu$ and $|\alpha| \leq 1 \Rightarrow \alpha\mathbf{x} \in B_\nu$. Also show that the origin lies interior to B_ν . Conversely, if $B \subseteq V$ is closed, bounded, convex, equilibrated subset with 0 in its interior, then a norm $\nu_B : V \rightarrow \mathbb{R}$ with unit ball B can be defined by $\nu_B(\mathbf{x}) = \inf\{\alpha > 0 \mid \frac{1}{\alpha}\mathbf{x} \in B\}$.

Solution

Claim 1: B_ν is closed, bounded, convex and equilibrated.

Closed:- Assume for the sake of contradiction, that there exist a sequence of points inside B_ν such that it converges to a value a , such that $a \notin B_\nu$. Therefore $\nu(a) > 1$, but since a is the limit of some sequence all of whose elements are in B_ν then $\nu(a) \leq 1$, which contradicts the assumption that $a \notin B_\nu$.

Bounded:- All points have norm ≤ 1 , and therefore it is bounded.

Convex:- Take any two points, $x, y \in B_\nu$.

All points $\alpha x + (1 - \alpha)y$, $\alpha \in [0, 1]$ are in B_ν .

Proof:- Since $\nu(\alpha x + (1 - \alpha)y) \leq \alpha\nu(x) + (1 - \alpha)\nu(y)$

$\nu(x) \leq 1$ and $\nu(y) \leq 1$, so $\alpha\nu(x) + (1 - \alpha)\nu(y) \leq \alpha + (1 - \alpha) = 1$.

So all points have norm ≤ 1 , and are therefore in the set. So B_ν is convex.

Equilibrated: For $|x| \leq 1$, $\alpha x \in B_\nu$.

$\nu(x) \leq 1$, $\nu(\alpha x) = |\alpha|\nu(x) \leq |\alpha| \leq 1 \Rightarrow \alpha x \in B_\nu$.

Origin lies in the interior: Take the open ball of radius $1/2$. This ball is contained inside B_ν , and since it is open all the points are in the interior.

Converse claim:- If B is a closed, bounded, convex, equilibrated subset with 0 in its interior then, a norm $\nu_B : V \rightarrow \mathbb{R}^+$ with unit ball B can be defined by $\nu_B(x) = \inf\{\alpha \geq 0 \mid 1/\alpha \in B\}$

Property #1: $\nu_B(x) = 0 \Leftrightarrow x = 0$

This is trivially true(atleast by setting $\nu_B(0) = 0$ in the definition).

Property #2: $\nu(\beta x) = |\beta|\nu(x)$

Lemma :- $\nu(x) = \nu(-x)$

Proof:- By equilibration, if $x \in B$, then $-x \in B$ as well.

Let $\nu(x) = \alpha_1$ and $\nu(-x) = \alpha_2$. If $\alpha_1 \neq \alpha_2$ then without loss of generality(WLOG) assume $\alpha_1 > \alpha_2$.

$1/\alpha_2 x \in B$, but this also implies $\frac{-1}{\alpha_2} x \in B$, so $\frac{1}{\alpha_2} x \in B$, but $\alpha_2 < \alpha_1$ and therefore contradiction, therefore $\alpha_1 = \alpha_2$.

Assume $\nu(\beta x) = \alpha \Leftrightarrow \frac{\beta}{\alpha} x \in B \Leftrightarrow \frac{\beta}{-\alpha} \in B \Rightarrow \frac{\beta}{|\alpha|} \in B$.

Property # 3: $\nu(x + y) \leq \nu(x) + \nu(y)$

$(\frac{1}{\alpha_1 + \alpha_2})(x + y) = (\frac{1}{\alpha_1 + \alpha_2})x + (\frac{1}{\alpha_1 + \alpha_2})y$.
 $(\frac{\alpha_1}{\alpha_1 + \alpha_2} \frac{1}{\alpha_1})x + (\frac{\alpha_2}{\alpha_1 + \alpha_2} \frac{1}{\alpha_2})y$ and since $\frac{\alpha_1}{\alpha_1 + \alpha_2} \frac{1}{\alpha_1} x \in B$ and $\frac{\alpha_2}{\alpha_1 + \alpha_2} \frac{1}{\alpha_2} y \in B$ then $\frac{\alpha_1}{\alpha_1 + \alpha_2} \frac{1}{\alpha_1} x + \frac{\alpha_2}{\alpha_1 + \alpha_2} \frac{1}{\alpha_2} y \in B$ because this is a convex combination of points in B , and therefore the norm has to be smaller or equal to $\alpha_1 + \alpha_2$.

5. The Hahn–Banach theorem says that for any norm ν on $\mathbb{C}^n$ and any subspace $V \subseteq \mathbb{C}^n$, if $\varphi : V \rightarrow \mathbb{C}$ is a linear functional whose maximum magnitude $|\varphi(\mathbf{x})|$ for $\mathbf{x}$ in the unit ν -sphere of V is α , then φ can be extended to a linear functional on $\mathbb{C}^n$ whose maximum magnitude on the unit ν -sphere in $\mathbb{C}^n$ is α . Show that the Hahn–Banach theorem is equivalent to the statement that separating hyperplanes exist for points outside of any closed, bounded, convex, equilibrated set that contains 0 in its interior.
6. Let V be the vector space of real (or complex) absolutely convergent series. The function $x \mapsto \|x\|_p = \left(\sum_{i=1}^{\infty} |x_i|^p\right)^{\frac{1}{p}}$ for $p \geq 1$ is a norm on V . (You may assume this, although no complaints if you'd like to prove it.) Prove that $\|x\|_p$ is not topologically equivalent to $\|x\|_q$ if $p < q$.

Solution

Assume $q > p$. If the two norms are topologically equivalent, then you can find a constant α and β such that $\alpha\|x\|_p \leq \|x\|_q \leq \beta\|x\|_p$.

Assume that this alpha does exist and is > 0 . We will make a sequence that has a sum of 1, and we will see what happens to the ratio when we take the number of terms to the limit.

Let $x_i = \frac{1}{n}$ for n terms and 0 otherwise

$$\text{Then } \sum_{i=1}^n |x_i|^p = \frac{1}{n^{p-1}} \text{ and } \sum_{i=1}^n |x_i|^q = \frac{1}{n^{q-1}}$$

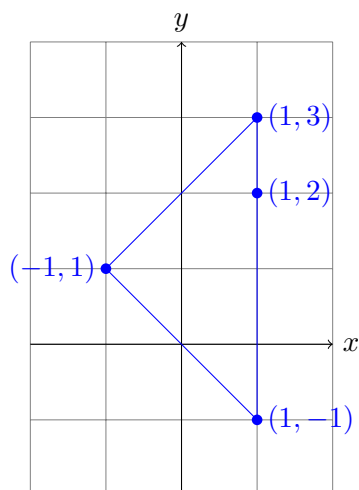
$$\text{Therefore } \frac{\|x\|_q}{\|x\|_p} = \frac{\left(\frac{1}{n^{q-1}}\right)^{\frac{1}{q}}}{\left(\frac{1}{n^{p-1}}\right)^{\frac{1}{p}}} = n^{\frac{p-1}{p} - \frac{q-1}{q}}$$

As $n \rightarrow \infty$, this ratio approaches 0 since $q > p$

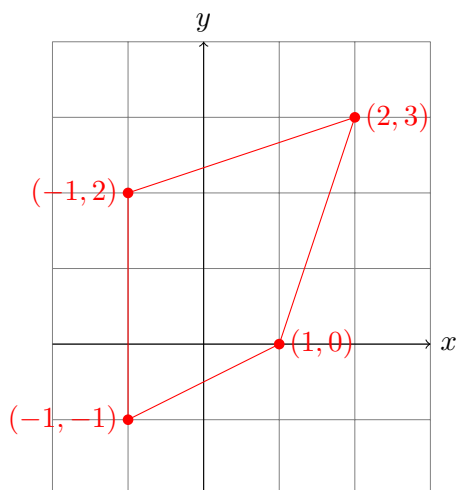
This contradicts the existence of $\alpha > 0$

7. Sketch the convex hull of $\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$ and $\left\{ \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$. /3

Solution



First Set



Second Set

8. Fix an operator $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^n$. If $X \subseteq \mathbb{R}^n$ is convex and $\mathbf{x}$ is an extreme point of X , must $\varphi(\mathbf{x})$ be an extreme point of $\varphi(X)$? What if φ is assumed invertible? /3

Solution

For the first part, the answer is no.

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

You can see that the matrix is singular and collapses the space to a line, so therefore the extreme points are no longer extreme points after the transformation. For example, the square with vertices $(0, 0)$, $(1, 0)$, $(0, 1)$, $(1, 1)$, the extreme points are $(0, 0)$, $(1, 1)$, $(0, 1)$, $(1, 0)$ but after the transformation the extreme points are $(0, 0)$ and $(0, 1)$.

For the second part, the answer is yes, because the transformation is invertible, therefore any line in the image can be mapped back to a line in the original space, if a point is an extreme point in X , and not extreme in the image, then it must be a convex combination of points in the output space, but since the transformation is invertible, you can take the inverse of the line to show that the point itself couldn't have been an extreme point, contradicting the assumption.

9. Prove that the intersection of finitely many closed halfspaces in $\mathbb{R}^n$ can have only finitely many extreme points. Must it have any extreme points at all? If it does have extreme points, must the intersection equal the convex hull of its extreme points? /3

Solution

Claim #1: Extreme points can only be on the boundary points of the intersection.

Proof: Assume for the sake of contradiction that there exists an extreme point x that is not on the boundary of the intersection. Then x has an open ball around it that is contained in the intersection. Then you can take two points in the open ball, and take a convex combination of the two points, which contains x , contradicting the assumption that x is an extreme point. Therefore all points are on the boundary, and the points on the boundary are points where the inequality of the halfspace space becomes equality.

Claim #2: A point that satisfies less than n restriction (satisfying restrictions means that the point is on the boundary of the halfspace) can't be an extreme point.

Take all the restrictions it does satisfy, then put them as a system of equations, and solve for a solution. The null space of the matrix has dimension > 0 because the point satisfies less than n restrictions, and therefore you can find a point in the null space p , and you can make $x + p$ and $x - p$, and these two points will also satisfy the restrictions, and their linear combinations will also satisfy the restrictions, as well.

$Ax=b$, Where A is the set of restrictions, x is the point, and b is the vector of constants (the c values for the halfspaces).

$$\text{rank}(A) < n$$

$$\exists p \in N(A) \text{ } N(A) \text{ is the null space of } A$$

$$x + p \text{ and } x - p \text{ satisfy the restrictions}$$

$$\alpha(x + p) + (1 - \alpha)(x - p) = x \text{ for } \alpha = 1/2$$

Therefore the point is in between the two points, and therefore can't be an extreme point.

Claim #3: A point that satisfies n or greater number of independent restrictions (meaning that the row vectors signifying the restrictions span the entire ambient space) is an extreme point.

Assume the point x satisfies n or greater number of restrictions, then the null space of the matrix is 0, now for the sake of contradiction assume that the point is not an extreme point, then there exists two points y, z such that $x = \alpha y + (1 - \alpha)z$. But the

elements in the halfspaces when evaluated with with linear funtional corresponding to the halfplane must have $\geq c$ for some constant.

But if you turn one of these $\geq (f(y) \geq c)$ into $>$, then you get

$$f(x) = c = \alpha f(y) + (1 - \alpha)f(z) > f(x) = c$$

, which is a contradiction, therefore the end points must also satisfy the restrictions, for all restrictions. But this implies that the points are also solutions to the matrix, which contradicts the fact that the matrix has a null space of 0.

Therefore the points satisfying n or greater number of restrictions are extreme points, and for any set of n restrictions there is only one point that satisfies all of them, and therefore the number of restrictions is bounded by $\binom{\text{total number of restrictions}}{n}$, which is finite, and therefore the number of extreme points is finite.

It is not necessariy that you have extreme points, if you have less restricitons than the ambient space, there are no extreme points for example if you have 2 halfspces in $\mathbb{R}^3$, and the intersection need not be equal to the convex hull, for example if you only have one extreme point, then the intersection isn't bounded.

10. Fix a column vector $\mathbf{b} \in \mathbb{R}^n$ and an $n \times n$ real matrix A . Show that the set of solutions [/3](#)
 $\{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} \leq \mathbf{b}\}$ of the inequality $A\mathbf{x} \leq \mathbf{b}$ is convex, where $\mathbf{a} \leq \mathbf{b}$ if $a_i \leq b_i$ for all i .
What kind of convex set is it? (No need to provide full justification of this last bit.)

Solution

Since you can write the inequality of each entry as a halfspace, and halfspaces are convex, then the intersection of the halfspaces is convex. The set is a polyhedron.

11. A *cone* is a subset $C \subseteq \mathbb{R}^n$ closed under nonnegative scaling: $\mathbf{x} \in C$ implies $\alpha\mathbf{x} \in C$ for all $\alpha \in \mathbb{R}_{\geq 0}$. A ray $\mathbb{R}_{\geq 0}\mathbf{x} = \{\alpha\mathbf{x} \mid \alpha \in \mathbb{R}_{\geq 0}\} \subseteq C$ is *extreme* if $\mathbf{x} \in \mathbf{y} + C \subseteq C$ implies $\mathbf{y} \in \mathbb{R}_{\geq 0}\mathbf{x}$. Let ν be an outer support vector for a closed convex cone C , so $\langle \nu, \mathbf{x} \rangle \leq 0$ for all $\mathbf{x} \in C$. Assume ν attains a unique maximum on C . Prove that the minimum angle between ν and a unit vector $\mathbf{x} \in C$ occurs when $\mathbf{x}$ lies on an extreme ray of C .

Solution

Claim #1: A point that achieves the minimum angle can't be in the interior of the cone.

Assume that the point that achieves minimum angle is a point in the interior. Then there is an open ball around it, that is contained in the cone, where all the points in the open ball have a greater angle than the point, but this implies that the point is in the same direction as the ν , because if you take that the plane that contains ν and the point, the only way for the angle to increase by moving both clockwise and anti-clockwise, is if the angle between them is 0.

Claim #2: Points on the extreme ray, are not 'movable in all directions(don't have an open ball around them)'.

Take a point on the extreme ray, assume that there an open ball around it where all points are inside the cone. Then take y a point that is not a linear scaling of x .

$$\begin{aligned} \text{Let } N &= x - y \\ x &\in N + C \\ x &= N + y, y \in C \end{aligned}$$

But since y isn't a linear scaling of x , that contradicts the definition of the extreme ray. Therefore x wasn't a point in the extreme ray, contradicting the assumption that you could make an ball around points on the extreme ray.

Since all points that have open balls around them can't achieve minimum angle, and only points on the extreme ray don't have open balls, the unit vector x has to be on an extreme ray, to achieve minimum.

12. How many $k \times n$ matrices of rank k with entries in a field $\mathbb{F}_q$ of size q are there? /3
Hint: how many nonzero row vectors are there in $\mathbb{F}_q^n$? How many vectors are linearly independent from the first one? How many are independent from the first two?

Solution There are $q^n - 1$ valid row vectors, for the first one, because there are q options for each position and excluding the zero vector, there are $q^n - 1$ options.

For the second one you have $q^n - q$, because you can't include all the linear scaling of the first vector which includes the zero, vector.

For the l th row, you have $q^n - q^{l-1}$ options, because there are previously $l - 1$ vectors and you have to subtract all the linear combinations from the set of possible vectors.

So the total number of rank k matrices is $\prod_{i=0}^{k-1} (q^n - q^i)$.

13. Show that the action of $GL_k(F)$ on the set of $k \times n$ matrices of rank k is *free*: if $g \neq g'$ are invertible $k \times k$ matrices and A is a $k \times n$ matrix of rank k , then $gA \neq g'A$. Use this (and the previous exercise) to compute the cardinality of the set $G_k(\mathbb{F}_q^n)$ of k -planes in $\mathbb{F}_q^n$. Hint: use #12 twice, once with $k = n$ and once with $k < n$.

Solution

Assume for the sake of contradiction that $gA = g'A$. The $gA - g'A = 0$, but $g - g' \neq 0$ and $A \neq 0$, but this implies that there is a non-zero row in $g - g'$ that is perpendicular to all the columns in A , but that implies all vectors are in the left null space of A but the left null space of A is 0, because the rank of A is k , which is the entire output space. So there can't be any elements to it, contradicting the fact that a nonzero vector is $g - g'$.

Since two different matrices in echelon form can't describe the same subspace, then the cardinality of $G_k(\mathbb{F}_q^n)$ is the same as the number of different echelon forms with dimension k . Given an echelon form you can use $GL_k(F)$ actions to arrive at different matrices, that have the same column space, so the number of different k planes is the number of distinct matrices divided by the number of number of matrices that describe the same subspace, which is the number of $k \times k$ invertible matrices.

So the cardinality is $\frac{\prod_{i=0}^{k-1} (q^n - q^i)}{\prod_{i=0}^{k-1} (q^k - q^i)}$.

14. Does the first claim in #13 remain true if the rank of A is less than k ?

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Solution

No, because if the rank of A is less than k , then there are non-zero vectors in the left null space of A . For example what if $A = 0$, then obviously $gA = g'A$ for all g, g' .

15. Use #13 to count the number of Sets in the game of Set. (Note: a Set is an affine line, [/3](#) not a linear subspace of dimension 1, so this is not quite a special case of #13. There is more than one way around this; I'll leave it to you to find at least one.) Confirm your count by counting a different way: a line L is determined by any pair of points on L .

Solution

The number of one dimensional subspaces is $(3^4 - 1)/(3 - 1) = 40$, therefore the number of lines that pass through the origin is 40, to get the total number of lines, move the starting around to all points, and since each line intersects three points, you will triple count each point, so divide by 3, so the total comes to $81/3 * 40 = 1080$.

Counting in a different way, a set is determined by any pair of points, the number of pair of points is $81 \times 80/2 = 3240$, and each line is counted by 3 different pairs, so you get $3240/3 = 1080$.